

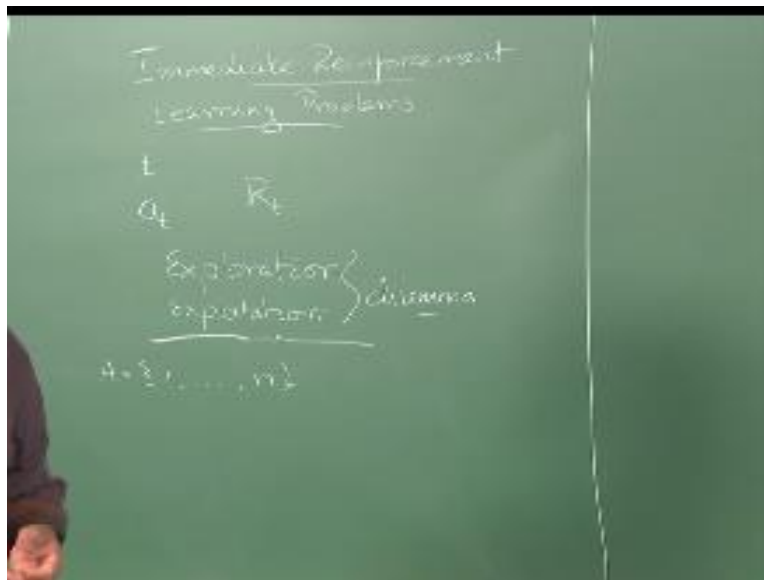
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REINFORCEMENT LEARNING

Introduction to Immediate RL

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So what we will start looking at today is what.

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Right so what we call immediate RL immediate reinforcement learning problems. So if you remember I said there are many, many characteristics of a reinforcement learning problem and one of those was that you get this evaluation from the environment. It could potentially be delayed right so I said you might actually lose a piece very important piece in the middle of a chess game

but you get a reward only at the end it is not like the final move you made is the one that causes you to lose it could be somewhere along the way right.

So what we are going to do now is to actually consider a much simpler version of the problem where we do not have this, the temporal nature to it right, so we are not we are not going to assume that the rewards are the scalar evaluation that comes from the environment is going to come at a farther time than when the action is actually performed right so I am going to assume that as soon as you perform an action like you are going to get a evaluation right and that is it basically right.

So we are going to consider a much more simplified version of the problem that is at every time instance I am required to pick an action right, so at every time instant t right I pick an action A_t right, and then I receive a reward R_t and that is it. my interaction is ended right. So I can learn something from this interaction and go back and pick another action at time $t + 1$ right so that is why we call this immediate reinforcement learning problems because the outcome is immediate so as soon as I take an action I get a reward right.

So what else is missing in my description here there is no real input to the system right there is no state signal that the system is seeing so all the input that it is getting are the rewards right there is no state input to the system right, so it is like it's just solving the same problem again and again right so instead of thinking of it as not having a state you could think of it as having like every time I give it the same input and I am asking you to give me a action in response to that does this seem like a reasonable thing to have can you think of cases where this might actually be an appropriate model.

So one example which people give is in drug testing right so I have I have a certain symptom that is presented to me right and I can choose on one of four different drugs to treat that symptom right so I first I give drug one it may or may not cure the person if it cures the person then I get a reward if it doesn't cure a person I do not get a reward right but it could very well be that for a variety of reasons the person did not get cured it is not because that the drug one was not the right drug to give for this person right. For this symptoms but for a variety of other reasons the patient might not have been cured so I cannot conclude by just administering the drug once that the drug is

ineffective for a set of symptoms right I have to keep trying the drug multiple times right before I can come to a conclusion that the drug is ineffective for a person correct right so for a certain set of symptoms so it is like saying that okay there are multiple you know samples that I have in a lab and each of them is exhibiting the same disease.

Let us not talk about human beings administering drugs and killing them or something can think of what samples in the lab right I'm going to add some drugs to that and see if that particular organism dies right in that in the when I add the drug right so it is essentially the same presentation as far as the drug testing is goes right it is the same organism that I am trying to kill say microorganism that I am trying to kill across all of these different instances it is like the same state that is being presented to me over and over again but then I have multiple choices that I could make and as soon as I do the choice I get a feedback saying okay.

Yeah the organism died or it didn't die right, so I have to keep trying this multiple times before I figure out which is the right drug to administer okay good is it clear so that gives you an example right so there are a couple of things which I kind of rolled into that example as well so one of those is that the outcome can be stochastic right, so every time I take an action right so I need not be given the same evaluation back right so I take a tissue sample add a drug to it right so i might not succeed every time right so sometimes I will succeed sometimes I might not succeed so there is some kind of a stochastic nature to the outcome right.

So as a consequence of the stochastic nature what did you have to do I had to repeat things multiple times right so this is these are things which you have to keep in mind when we are talking about this kind of immediate reinforcement learning problems that we are going to have stochastic outcomes and then we have to think about repeating choices multiple times before we can be sure of the outcome right is there anything else that was that would have been critical to the successful outcome of this let us say that I tried drug one okay.

And the organism survived okay, so then I just can I keep using drug one for all the other organisms as well let us say let us keep the problem simplified right I just will say have only two drugs drug one drug two okay I take the first organism give it drug one it survives I take the second organism

give it drug two does not survive okay, now can I conclude that which one is more effective drug one or drug two we don't know I am assuming is ones is survival what we are looking for not survival well what we are looking for not survival talking about microorganisms here right I actually flipped it around but nobody actually noticed it.

So yeah so I am looking for non survival of the microorganism right so in this case drug two appeared to be more effective than drug one right so that does not mean I can keep giving drug two to all the other samples right I need to actually go back and try drug one a few times right so this is the exploration right I was telling you about exploration in the previous class right this is the exploration here that I have to do exploration so the critical thing here that even the simple case of this immediate reinforcement learning introduces to is the need for exploration right but can I just keep exploring can I just keep giving random drugs to each of those samples right at some point I have to stop right.

I am interested actually in killing those microorganisms right at some point I have to stop and decide okay here this is the actual drug that is going to kill the microorganism and I have to start giving that drug right this need not be an explicit point where I stop we will see as we go along but I am interested in eventually exploiting the knowledge that I am acquiring about this the whole system right, so either I explore or I have to at some point I have to stop and start exploiting so there is this inherent tension between exploration and exploitation okay.

They are in some sense at loggerheads with each other right if I stop at some point and say okay I have know enough about the system and then I start exploiting then I necessarily have to stop my exploration right but then if I do not explore enough I will be exploiting a poor solution right so this exploration exploitation so sometimes people call it a conflict sometimes it is called the called the Explore exploit dilemma right so should I explore or should I exploit so that is essentially what this immediate reinforcement learning problems captures for us okay.

Since this is a very crucial question in reinforcement learning so we will look at this immediate reinforcement learning problems in a little bit of detail so that it will motivate some of the other things that we will do later when we move to the full RL problem right and so what I will do which

is slightly different from what is there in the textbook, so how many of you have actually looked at the textbook good see I did not ask how many of you have read the book right. So what I will do is slightly different from the book and I am still not decided whether it will be for both versions of the course let us talk a little bit more about some of the theoretical questions that people are interested in addressing in the immediate RL setting right.

The book just gives you a very very algorithmic very prescriptive viewpoint of this right this is oh here is how you solve the immediate reinforcement learning problem right it does not go beyond that it does not give you anything about right so why do people solve it the way they do it right I will go a little bit beyond that and try to give you some intuition in that mainly because that seems to be a very active area of research nowadays in reinforcement learning and many of the ideas that people are using there are also being transferred to the full reinforcement learning setting.

The one with all the delayed rewards the states and things like that right so lot of ideas from the research in this immediate RL setting is also being used in the full RL setting so I want to introduce you to those topics right, yeah so despite the makeup of the class right which is predominantly undergrad this is a graduate level course right so I want you to want to introduce people to research questions and other things there have been demands for I to actually teach an advanced RL course at some point I know at least one person who'll take it I am going to try this.

Now every class I am going to make people turn and look at you at least once I managed to do that last class anyway so go back so we will look at a little bit of formalism on this right so what I am going to assume is that I have a set of actions A which I will index by 1 to n so action one could be anything right and I am just giving arbitrary numbers 1 to n it could be drug one drug two drug three right or it could be ad one ad two ad three whatever it is just 1 to n is what we will denote it as right.

And each action when I pick that action right gets a payoff it is an evaluation so I am going to be using these terms interchangeably right reward payoff evaluation cost so all of this denote the same thing that dotted line that came in my box diagram right so depending on what literature you come from you use different different names for it right so the reward typically comes from guess

psychology right cost comes from control theory optimization yeah payoff comes from economics and what is the last thing I said evaluation comes from optimization okay fine.

So depending on which, which background you come from you give it different names right and so I will use them interchangeably for example for most of this literature I will be using payoffs because a lot of the work has been done in the in the OR and Economics community in when they are trying to analyze these kinds of problems right so when you pull each when I actually select an arm I select an action right what's going to happen is that I will get a reward or a payoff sampled from some unknown distribution so what do I mean by that stochastic but what do I mean more I mean it can right this is something which I found people have difficulty so some people at least have difficulty when I say it is sampled from a distribution so what I mean by mean is the following.

Let us keep it simple so that people can conceptualize it so associated with each action let us say that I have a coin associated with each action there is a coin okay that has some probability of coming up heads and some probability of coming up tails okay so let some probability P of coming up heads lets say I pick action one then there is a probability P_1 of coming up heads when I toss the coin right and I will say that if the coin comes up heads ok I will give you a payoff of 1 if the coin comes up tails I will give you a payoff of 0 right.

So every time I select an action I am going to toss this coin so depending on whether it comes up heads or tails I will give you either a one or a zero right so the payoff that I will get every time I select that action is not the same right but what is same the coin right if I select actual action one I will always toss coin one if I select action 5 and always toss coin five right I am not changing these coins okay, is it clear does it make sense to people right I will not change the coins the coins will be the same right.

But every time I pick an action I will toss that coin and I will give you the payoff or reward corresponding to the outcome of the coin toss okay so if I say that the probability of reward for picking action 1 is 0.8 what does it mean the probability of heads of the associated coin is point eight right in this case I can go further I can also say something else I can say that the expected

payoff of picking action one is point eight as well right because the probability point eight I will get a reward of 1 with the probability point 2 I will get a reward of 0 so if I take the expectation of this it will get will be point eight right.

So people understand this now so when I say select an arm select an action I will tell you why I keep calling arm in a second and then I will keep calling it arm so when I select an action right then I am going to toss a coin right depending on the outcome of the coin I am going to give you reward this is something which you should remember when you are actually going and implementing these things right so I had people do all kinds of weird things right and make sure that you remember what I mean when I say you sample from a distribution okay great.

So now I gave you the simple example of tossing a coin I could actually have any complex distribution right so if a toss a coin what is the distribution I am sampling from? Bernoulli right I could have other kinds of distributions I could have a Gaussian distribution let us say so now what does it mean for every action I choose okay.

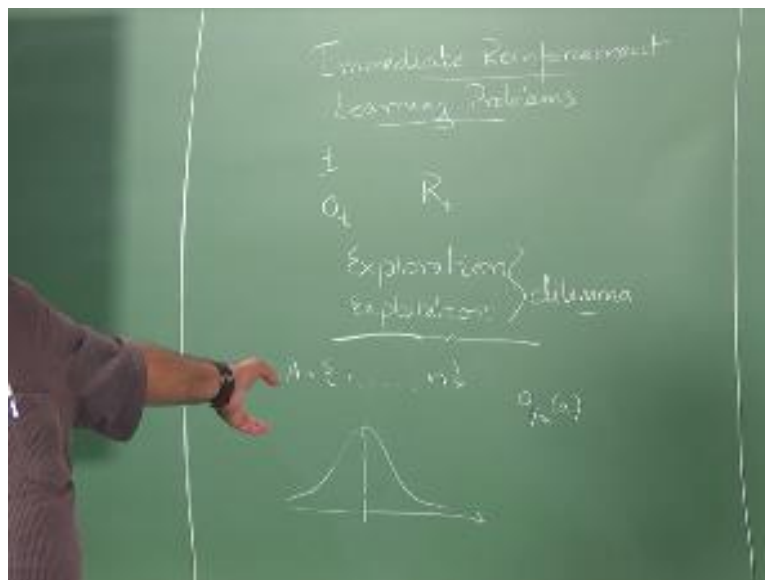
So for action 1 there is going to be a like some kind of a Gaussian distribution right and so what is the axis here x axis here the payoff the payoff or the reward x axis is the payoff or the reward and the y axis is the probability that you will see such a payoff right so that is the most probable payoff as it also happens to be the for Gaussians this also happens to be the expected payoff right so how will I sample from a Gaussian pick a random number and 0 and anyone else how do I sample from a Gaussian okay.

How do you do that people who are in my minor I told you how to do this in the class okay little bit more efficient way of doing it yeah and then what you do? Do not tell me you use Matlab no man it is you know too many terminology you just keep throwing them around what does MC stand for in, Markov Chain, then there are two MCs then you just told me a Monte Carlo I said give me something more efficient and then you keep saying another Monte Carlo method what is this no one oh come on, fine okay Pallavi you tell me, so the guys who were in the you guys were there in my minor thing right I actually told you how to sample do you remember I said so I did

this with a discrete distribution and then I told you how to do this with continuous distributions as well can you give them a write-up on moodle okay.

So basically the idea is you generate a random number right between 0 and 1 generate a uniform random number between 0 and 1 right and then keep integrating the area under the curve what is it called the CDF right and then when the CDF reaches that random number you generate so that is the number you sample does it make sense, so this is a complete aside.

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Let us say that I have four possible outcomes right I want to generate one of those four outcomes at random right so the probabilities are p_1 p_2 p_3 p_4 how will I do that I will generate a uniform random number between 0 to 1 right, now so I will see if the number is less than P_1 if the number is less than P_1 I will generate outcome 1 if the number is between p_1 and p_2 and generate the outcome of two this number is between p_2 and p_3 I generate the outcome of three generate number is between p_3 and p_4 is it correct or wrong, it should be between p_1 and p_1 plus p_2 okay it should be between $p_1 + p_2$ and $p_1 + p_2 + p_3$ so it should be the cumulative sum right it should be the cumulative sum previous cumulative sum and the next cumulative sum.

So if it lies between that then that is the outcome right so if it is less than P_1 I will generate one if it is between p_1 and $p_1 + p_2$ I will generate two if it is between $p_1 + p_2$ and $p_1 + p_2 + p_3$ I will generate three if it is between $p_1 + p_2 + p_3$ and 1, I will generate four right do not tell me if $p_1 + p_2 + p_3 + p_4$, so you can simulate the CDF for the Gaussian right yeah so there are ways of doing it right so there we have doing this so you can do this for any probability distribution for which you can define the CDF.

Not just Gaussian it's not just discrete so the discrete case it is very clear right so what you are doing is essentially adding up the probability so whenever you hit that number you say that is the sample that you have drawn right in this case you just do this for the Gaussians okay so this is what I mean when I say sample from a probability distribution again it is not the most probable outcome like when I say sample from this Gaussian I have actually seen people implement okay that's the reason I keep reiterating this people actually form the Gaussian and then keep returning the value that has the highest probability right.

So please do not do that of course if you are using MATLAB or something like that you don't even have to think about it we can just say `rand N` and then it will give you a sample from a normal distribution which is what Subho was going to tell you is the way to sample from a Gaussian but this is what it does internally right and to give you the sample having said this now you can go back to using `randn` yeah you had a question okay fine so going back associated with each of these actions right I am going to have some distribution from which the payoff is being sampled okay is it clear.

And these distributions I am going to assume at least initially will be fixed for the duration of the experiment for duration of the learning this distributions will be fixed like I said you do not change the coin you don't change the Gaussians also okay now if I knew the Gaussians or if I knew the coin if I knew the probability with which the coin will turn up tails heads right so how will I solve the problem there is no problem right I just picked the action that has the highest mean right highest mean and for the Gaussian or the highest p_1 for the coins right whatever it is the highest expected payoff I just picked that arm picked that action right.

So the challenge comes when I do not know the distributions all I know is there is some distribution which I am assuming is fixed according to which the payoffs are generated I do not know the distribution so I basically have to figure out the distribution and then pick the action I picked action corresponding to that right so is that clear so for each of these actions right for each of these actions I am going to assume that there is a true expected payoff which we will denote by $q^*(a)$.

Let us say it's action so q^* star is a function that will give me the true expected payoff of action a right so in the case of the coins $q^*(1)$ will give me p_1 which is the probability of heads right $q^*(2)$ will give me p_2 and so on so forth in the case of Gaussians $q^*(1)$ will give me μ_1 right $q^*(2)$ will give μ_2 and so on so forth so that's essentially what the q^* function is this is again unknown to me right if I know q^* I am done right I mean the problem is solved right but this is the assumption we are going to make so we have a set of actions A and we have a set corresponding q^* that we do not know about right.